



Eidgenössische Technische Hochschule Zürich
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Deep Learning-Based Segmentation of Female Pelvic Organs and Endometriotic Lesions in Ultrasound Images and Videos

Bachelor Thesis

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Abstract

Endometriosis is a chronic condition affecting approximately 10% of reproductive-age individuals assigned female at birth, often leading to severe pain and infertility. Despite its prevalence, diagnosing endometriosis remains challenging, with delays ranging from 4 to 11 years due to the reliance on invasive laparoscopic procedures. This thesis explores deep learning-based approaches to improving the segmentation of pelvic organs and endometriotic lesions in ultrasound images and videos, aiming to enhance non-invasive diagnostic capabilities.

Building upon an existing 2D U-Net model developed by Scanvio, multiple enhancements were investigated. These included the use of Masked Autoencoders (MAE) for self-supervised feature extraction, the incorporation of temporal context via 3D and hybrid 2.5D convolutions, and adjustments to loss functions to counteract class imbalance. The best-performing approach involved pretraining a Masked Autoencoder and using the learned encoder for feature extraction, which led to an 18.55% improvement in lesion segmentation performance over the baseline model. Other methods, such as full 3D convolutions, did not yield significant improvements and were limited by computational constraints and dataset size.

These findings suggest that self-supervised learning techniques like Masked Autoencoders can significantly improve deep-learning-based segmentation of endometriotic lesions in ultrasound images. Future research could explore transformer-based architectures, 3D Masked Autoencoders, and temporal consistency losses to further enhance performance and reliability.

Contents

Contents	iii
1 Introduction	1
1.1 Background and Motivation	1
1.1.1 Understanding Endometriosis	1
1.1.2 Challenges in Diagnosing Endometriosis	2
1.2 Research Objectives and Contributions	2
1.3 Outline of the thesis	3
2 Related Work	5
2.1 Deep Learning for Endometriosis Detection in Ultrasound . .	5
2.2 U-Net Architecture	6
2.2.1 3D U-Net	6
2.3 Masked Autoencoders	8
3 Method	9
3.1 2D Convolution on Single Frames (Baseline U-Net)	9
3.1.1 Adjusting Loss Weights for Class Imbalance	13
3.2 Pretraining a Masked Autoencoder for Feature Extraction (Pre- trained U-Net)	14
3.3 2D Convolution on Image Sequences (Channel-3D U-Net) . .	16
3.4 3D Convolution on Image Sequences (3D U-Net)	18
3.5 Hybrid 2D/3D Convolution on Image Sequences (2.5D U-Net)	20
3.6 Addressing Determinism in Experiments	21
4 Results	23
4.1 2D Convolution on Single Frames (Baseline U-Net) And Ad- justed Loss Weights	23
4.2 Pretraining a Masked Autoencoder for Feature Extraction . .	23
4.3 2D Convolution on Image Sequences	25

CONTENTS

4.4	3D Convolution on Image Sequences	26
4.5	Hybrid 2D/3D Convolution on Image Sequences	26
4.6	Overall comparison	26
5	Discussion and Conclusion	29
5.1	Summary of Key Findings	29
5.2	Limitations and Challenges	30
5.3	Future Research Directions	30
	Bibliography	31

Chapter 1

Introduction

Endometriosis is a condition affecting around 10% of women and those assigned female at birth that are of reproductive age [1]. It is defined by endometrial tissue growing outside of the uterus, and symptoms range from pain all the way to infertility [2]. It is often difficult to diagnose in ultrasound, which is why the current standard for diagnosis is an invasive laparoscopic surgery. This is part of the reason why diagnostic delays tend to be large, ranging from 4 to 11 years [3].

The basis of this thesis was a 2D convolution U-Net model developed by Scanvio that was trained to segment female pelvic organs and endometriotic lesions in pelvic ultrasound images. The goal of this thesis was to improve the segmentation quality by expanding and adapting the existing model.

1.1 Background and Motivation

1.1.1 Understanding Endometriosis

Endometriosis is a disease in which tissue similar to that of the lining of the uterus, called the endometrium, grows outside of the uterus and around the pelvic area [4]. This tissue continues to follow and react to the menstrual cycle, thickening, breaking down, and bleeding. However, the tissue is trapped within the body as it cannot exit through the vagina like the lining of the uterus does. This can lead to inflammation, scar tissue formation, and adhesions. Such adhesions may lead to pelvic tissue and different organs becoming stuck to one another, causing further complications.

Those with endometriosis may suffer from pelvic pain, pain during intercourse, pain with bowel movements or urination, excessive menstrual bleeding, and even infertility [2]. Other common symptoms include fatigue, nausea, diarrhea, constipation and bloating. Due to the nature of the disease, most symptoms usually appear or are worsened during menstrual periods.

Unfortunately, endometriosis is not a rare disease by any means: Approximately 10% of women and those assigned female at birth that are of reproductive age suffer from it [4, 5, 1, 6].

1.1.2 Challenges in Diagnosing Endometriosis

Despite this, endometriosis has historically seen a lot less research funding when compared to other diseases of similar prevalence and impact, as highlighted by Kirk et al. [7]. In the US in 2022, diabetes had received 15 times more budget per female patient than endometriosis, while Crohn’s disease, an inflammatory bowel disease, received over 65 as much budget per patient.

The delay between symptom onset and definitive diagnosis can range from 4 to 11 years [3]. The reasons for this are manifold [8]: For one, the current gold-standard for diagnosing endometriosis is through visual inspection by means of laparoscopy. Given that this is a surgical procedure, it is not a step taken easily or early in the diagnostic process. For another, many symptoms have been normalized to simply be a part of menstruation, making it less likely for both healthcare professionals and those affected to take them seriously. Lastly, due to the varied nature of symptoms, misdiagnosis is a common issue. For example, endometriosis is sometimes mistaken for the aforementioned Crohn’s disease [9].

Pelvic ultrasound could present an efficient and low-cost method to detect endometriosis, but it requires sufficient expertise and knowledge by the operator in order to be effective [10]. A study by Zhang et al. [11] found that no non-invasive diagnostic method currently available is good enough to replace laparoscopy as the standard for diagnosing deep-infiltrating endometriosis.

1.2 Research Objectives and Contributions

This thesis is part of a larger research effort undertaken by the ETH spin-off Scanvio in cooperation with the Medical Data Science Group from ETH. Scanvio’s goal is to enable ultrasound operators to benefit from expert-guided ultrasound using the help of AI. Scanvio has developed an ultrasound image segmentation model with the goal of accurately detecting and localizing both organs in the pelvic region as well as endometriotic lesions. The aim of this thesis was to explore ways to improve upon this model.

Organs like the uterus tend to maintain a uniform shape across various frames, which is why a single frame can be enough to accurately identify those structures. Lesions on the other hand do not follow such a pattern.

Their appearance will vary much more: not only between frames of the same video, but also from patient to patient.

When looking for lesions in ultrasound videos, doctors would often scroll between adjacent frames to detect such inconsistencies. That is why our first idea was to give the model additional context by switching from 2-dimensional convolutions on single frames to 3-dimensional convolutions on multiple frames that relate temporally and spatially.

Our second idea involved getting the model to learn more robust representations that do not solely rely on labels. To that end, we performed unsupervised pre-training of a Masked Autoencoder to be used for feature extraction. The resulting encoder would then be used to finetune a model on the downstream segmentation task. Doing this also allowed us to make use of a large volume of unlabeled data along with the much smaller labeled dataset.

1.3 Outline of the thesis

In the introduction, a brief overview is given of the disease, its relevance, and issues with the current diagnostic practices. The related work section starts with existing deep-learning approaches to detect endometriosis using ultrasound, followed by an overview of the main deep-learning models that this thesis builds upon. The methods chapter describes the conducted experiments as well as their setup. The outcomes of the experiments are then reported on in the results chapter. The last chapter—the discussion—recaps key findings, reflects on the experiments and gives ideas for further research.

Chapter 2

Related Work

The first section will briefly present existing work that tries to detect endometriosis in ultrasound imaging using deep learning. The sections after that will introduce the main methods this thesis builds upon: Section 2.2 focuses on the original U-Net architecture, while section 2.2.1 is about 3D and 2D/3D hybrid adaptations of the U-Net. Section 2.3 is focused on the Masked Autoencoder.

2.1 Deep Learning for Endometriosis Detection in Ultrasound

Endometriosis affects the quality of life of so many women and those assigned female at birth around the world [4, 5, 1, 6]. The delay from symptom onset to diagnosis can be painstakingly long [3, 8, 7]. And yet, few studies have investigated the automatic detection of endometriosis through ultrasound imaging with the help of deep-learning [12].

Maicas et al. [13] developed a ResNet based deep-learning model to accurately (AUC of 0.96) detect Pouch of Douglas (POD) obliteration from ultrasound videos of the sliding sign test. The Pouch of Douglas is a small cavity between the uterus and the rectum. POD obliteration refers to the complete or partial closure of this space, which can occur due to adhesions caused by endometriosis. The sliding sign test checks for smooth motion of the front wall of the rectum against the back of the cervix and uterus. A negative sliding sign suggests that something is hindering smooth motion, indicating POD obliteration and with that possibly endometriosis [14].

In a proof-of-concept study, Balica et al. [15] explore the effectiveness of five different model architectures (XCeption, Inception-V4, ResNet50, DenseNet, and EfficientNetB2) for detecting endometriosis in trans-abdominal ultra-

sound videos. The highest performing model was DenseNet with an AUC of 0.9.

According to Podda et al. [16], their work is the first in literature trying to utilize deep-learning to segment trans-vaginal ultrasound images in relation to endometriosis. They proposed a multi-scale approach: The same model was trained multiple times, each time on a different image resolution. The predictions from different scales were then combined to form a final prediction mask. They did this with three different models, namely DenseNet121, VGG19, and a custom U-Net based model developed by them. Their best performing model was DenseNet121, with a dice score of 0.818. It is important to note that the models were only trained on images containing lesions, i.e. positive samples.

2.2 U-Net Architecture

The U-Net architecture, first introduced in 2015 by Ronneberger et al. [17], is a highly efficient approach to training segmentation models when only little annotated data is available. The architecture contains a downsampling encoding path and an upsampling decoding path. On a graph this results in a U-shape, hence the name. On the encoding path, features are extracted while successively being downsampled. On the decoding path, low resolution features are successively combined and the output is upsampled.

To help create high resolution outputs, skip connections between layers of the same height in the downsampling and upsampling path are implemented. These skip connections concatenate higher resolution features extracted on the downsampling path with outputs on the upsampling path. Combining these two allows U-Net models to generate accurate segmentation outputs of the same resolution as its input.

Training data augmentation (such as elastic transformation) is used to make the available training set appear larger than it is. Due to the nature of biomedical images, elastic transforms thereof typically still result in realistic samples that could be encountered in practice. This, coupled with the fact that biomedical datasets are often only sparsely annotated, has lead the U-Net architecture to become very popular for use in the medical field [18, 19, 20, 21, 22, 23, 24, 25].

2.2.1 3D U-Net

In [26], Çiçek et al. adapt the original U-Net by Ronneberger et al. to work with 3-dimensional input and output. To that end, they replaced all 2-dimensional operations (i.e. convolutions and max pooling), and replaced them with their 3-dimensional counterparts. They tested the performance

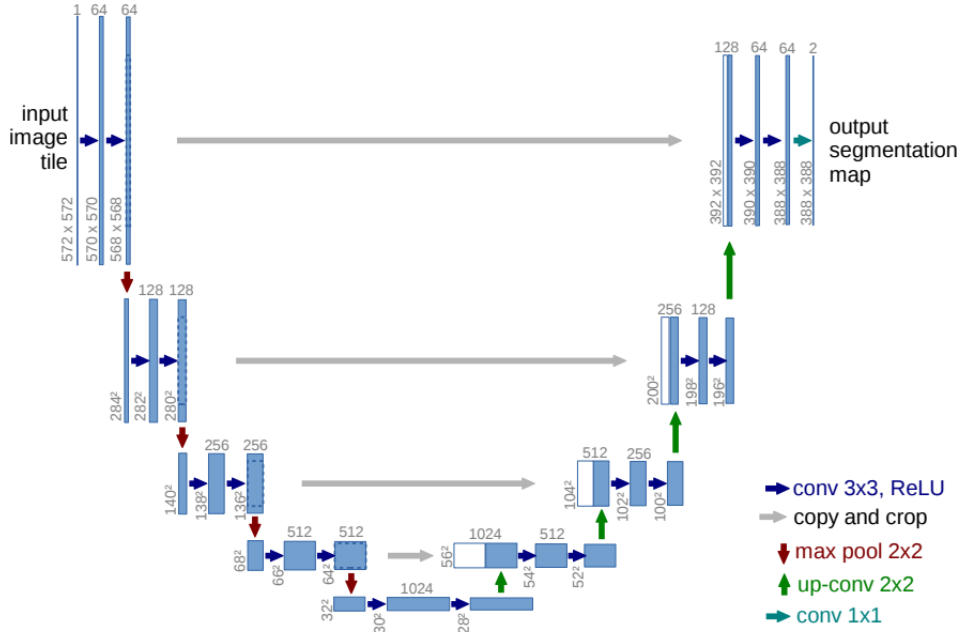


Figure 2.1: A diagram of the UNet architecture taken from Ronneberger et al. The caption reads: “U-net architecture (example for 32x32 pixels in the lowest resolution). Each blue box corresponds to a multi-channel feature map. The number of channels is denoted on top of the box. The x-y-size is provided at the lower left edge of the box. White boxes represent copied feature maps. The arrows denote the different operations.”

on the segmentation of a *Xenopus* kidney. Given volumetric data comprised of slices as input, with only a few of the slices being annotated, the model was able to generate a dense segmentation with an IoU of 0.863. Similarly, when trained on sparsely annotated samples, the model was able to segment a new, unannotated sample from a similar distribution with an average IoU of 0.723.

A method that combines both spatial information obtained from 3D convolutions and the increased computational efficiency of 2D convolutions was evaluated by Zhou et al. [27]. In what they call D-UNet, they first perform both 3D and 2D convolutions on the input in two separate paths in the downsampling layers. The output of the 3D path is then compressed into a single spatial dimension and combined with the features on the 2D path. The remaining downsampling paths and all upsampling paths only use 2D convolutions.

Using this hybrid architecture, they were able to achieve better results on their test data than with a purely 2D or 3D approach, while keeping model parameters and therefore required GPU memory low.

2.3 Masked Autoencoders

In 2021, He et al. published a paper titled "Masked Autoencoders Are Scalable Vision Learners" [28]. They describe how they made use of a training paradigm from natural language processing to pre-train vision transformers in a self-supervised manner. They feed a model images with various patches being masked, and train it to predict those masked patches. The masking ratio is set very high (75%) in order to prevent the model from being able to just connect lines and instead force it to gain some semantic understanding of the image. By having the encoder part of the model only process visible patches, their approach remains efficient at scale.

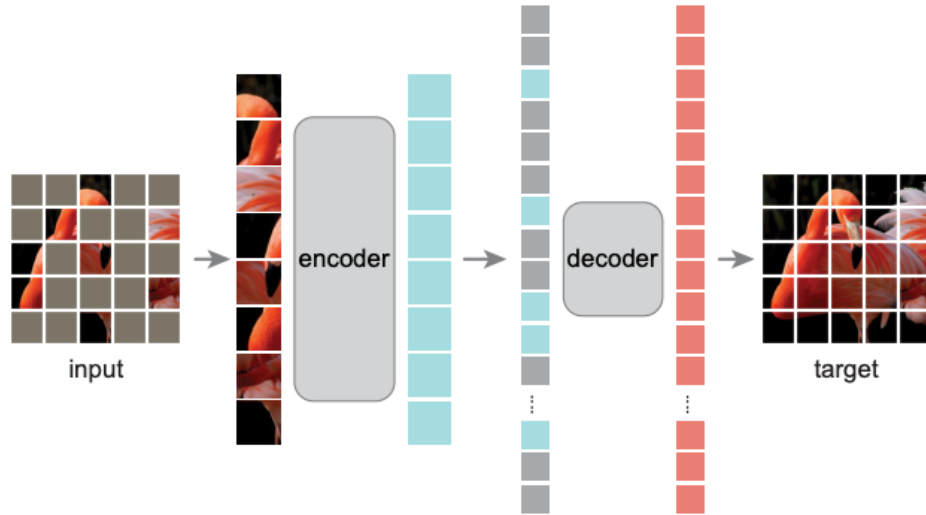


Figure 2.2: A diagram showing the MAE architecture taken from the paper by He et al. The caption reads: "During pre-training, a large random subset of image patches (e.g., 75%) is masked out. The encoder is applied to the small subset of visible patches. Mask tokens are introduced after the encoder, and the full set of encoded patches and mask tokens is processed by a small decoder that reconstructs the original image in pixels. After pre training, the decoder is discarded and the encoder is applied to uncorrupted images (full sets of patches) for recognition tasks."

Using this method, they were able to achieve better performance on downstream tasks such as object detection, semantic segmentation, and classification than with supervised pre-training methods. On the ImageNet-1K dataset, they surpassed previous state-of-the-art (SOTA) in classification accuracy.

Chapter 3

Method

All experiments were implemented using Python and the PyTorch and PyTorch Lightning libraries. All model training was done on a single NVIDIA RTX 4090 GPU, using BFloat16-Mixed precision.

3.1 2D Convolution on Single Frames (Baseline U-Net)

To establish a baseline for all experiments, the existing model that Scanvio had developed was trained and evaluated.

Dataset

The dataset used to evaluate the baseline (and most other) models contains 1550 labeled images across 317 videos and more than 120 patients. Most videos contain between 2 and 7 labeled frames (mean: 4.89, SD: 2.58). Each image in the dataset is labeled with segmentation masks for one or multiple of the following segments: uterus, vagina, bladder, and lesion. The vagina and the uterus are visible on almost all images. The bladder appears on about 2/5 of images, and lesions are present in a bit more than 1/10.

Table 3.1: Breakdown of segments by how many images they are present on.

Segment	Count	Percentage
Vagina	1537	99.16%
Uterus	1351	87.16%
Bladder	635	40.97%
Lesion	203	13.10%

A detailed breakdown of the images by the segments they feature is shown in the following table:

3. METHOD

Table 3.2: Breakdown of images by segments present. Each image is only counted once.

Segments present	Count	Percentage
Uterus, vagina	619	39.94%
Uterus, vagina, bladder	531	34.26%
Uterus, vagina, lesion	156	10.06%
Vagina	126	8.13%
Vagina, bladder	65	4.19%
Uterus, vagina, bladder, Lesion	35	2.26%
Other combinations (count < 10)	17	1.10%

5-fold cross validation was performed, so each test set contained 20% of the dataset. The remaining 80% were split into 64% training set and 16% validation set. Videos from the same patient could appear in both the training and the validation set. For the test set, a per-patient split was ensured.

Preprocessing

All images were resized to 384x256 using bicubic resampling [29] and converted to grayscale, with their aspect ratios being preserved using padding with black pixels if necessary. Images from the training set were further augmented using the Albumentations [30] library (version 1.4.11). Each time a datapoint was used during training, augmentations were applied with different probabilities. Image and segmentation masks received the same augmentations.

Table 3.3: Summary of used image augmentations

Transform	Probability	Effect
ColorJitter	0.5	Randomly applies brightness and contrast scaling within a defined range
ElasticTransform	0.25	Creates random elastic transforms that simulate soft tissue deformation
Affine	0.25	Applies geometric transformations with random parameters (within a defined range) for rotation, scaling, shearing, and translation
Flip	0.25	Flips the image horizontally
Blur	0.25	Applies Gaussian blur with random strength

Architecture

The model follows the U-Net architecture and has 5 layers each on the down-sampling and upsampling path. In each layer, the number of channels is doubled while the input dimensions are halved. This halving is implemented using a max pool operation with a kernel of size 2x2. After the last down-sampling layer, the feature map has a size of 12x8 with 1024 channels.

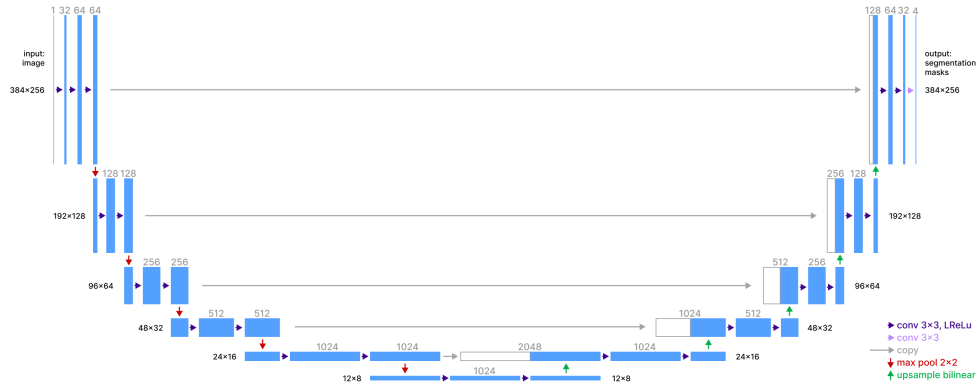


Figure 3.1: U-Net architecture of the baseline model. Each blue box represents a multi-channel feature map. The number of channels is denoted on top of the box. The x-y-size of all feature maps in a layer is denoted both all the way on the left and the right. White boxes are the copied feature maps. The arrows symbolize different operations.

Between the down- and upsampling paths we add two more convolutional layers with identical input and output sizes. The upsampling path is symmetrical to the downsampling path, halving the number of channels at each layer while doubling the output dimensions. This doubling is implemented using bilinear upsampling.

After the last upsampling layer, a final convolution reduces the number of channels from 32 to 4, one channel for each predicted segment.

All convolution operators use a kernel of size 3x3. Batch normalization is applied after every convolution except for the input and output convolution. Leaky ReLU [31] is used after every convolution except for the output convolution.

Training

One training epoch consisted of 200 batches with a batch size of 8. This made for a total of 1600 samples seen per training epoch. These samples were drawn from the dataset in random order, and all datapoints were shuffled every epoch. After every training epoch, a validation round was performed. One validation round would include each sample from the val-

3. METHOD

validation set once. We trained until we did not observe any improvement to the validation loss over 15 epochs.

We used a focal loss [32] with $\gamma = 2$ and $\alpha = 0.25$. The loss of a single sample was given by the sum of the losses of each segment (uterus, vagina, bladder, lesion) for that sample.

Focal loss downweights the loss of easily classified pixels to counteract the size imbalance between background and foreground. In our dataset, the foreground (i.e. segmentation masks) generally contained much fewer pixels than the background.

Table 3.4: Segment mask sizes as a ratio of the image size

Segment	Mean	Std Dev	Median
Bladder	0.05	0.06	0.03
Lesion	0.05	0.05	0.03
Uterus	0.12	0.07	0.12
Vagina	0.01	0.01	0.01

In addition to the loss of the final output, a deep supervision loss for the second to last 3 output layers was added, with lower layers contributing less. The output of each layer was convolved to reduce the channel size to 4, and then upsampled to match the output dimensions. The loss for each output was then calculated as described above and scaled down before being added to the final output loss. Starting with the second to last layer, the deep supervision losses were scaled by 0.1, 0.1^2 , 0.1^3 , and so on, respectively.

The Adam [33] optimizer was used with an initial learning rate of $1e^{-4}$. Additionally, the learning rate was halved whenever the validation loss did not improve over 5 epochs.

All models were trained on 5 different seeds: 55555, 66666, 77777, 88888, and 99999.

Evaluation

During training, the model was continuously saved at checkpoints. At the end, the checkpoint with the lowest validation loss was restored to be evaluated on the test set.

For each datapoint in the test set, various metrics were logged. The primary metric relevant for evaluation was the Dice score, which both measures the similarity between a prediction and the ground truth. A pixel was predicted to belong to a segment if the output probability for that pixel was greater

than 0.5. These output probabilities were obtained by applying a sigmoid to the grayscale output of the model.

For a select few images, we also logged the output probabilities for each segment along with the discrete outputs (i.e. probability > 0.5). The images were chosen such that they included easy, medium, and hard to predict cases from various videos. The idea was to see which predictions would change most across different experiments.

3.1.1 Adjusting Loss Weights for Class Imbalance

In table 3.1 it can be seen that our dataset included many more labels for the segments uterus and vagina than for bladder and especially lesions. In previous experiments we noticed that the prediction quality for these segments followed the same trend: uterus and vagina were predicted reasonably well, while the model struggled a bit with the bladder and a lot with lesions.

In the baseline model, the losses from all segments are simply summed to form the loss of a single datapoint. This means that for a single datapoint, all segments affect the loss equally. Because not all segments appear equally often though, not all segments will have contributed to the loss equally when looking at the entire training process.

In this experiment we tried to counteract this by scaling the loss of each segment such that each segment would contribute equally across the entire training process. All else remained just as described for the baseline (see chapter 3.1).

These loss adjustment weights were calculated as follows: we first obtain the unnormalized weight factors w_x by dividing the number of labels available for segment x by the total number of labels available (i.e. the sum of the counts in table 3.1). Each weight factor w_x is then normalized such that their sum remains equal to the number of segments:

$$w'_x = \frac{w_x \cdot |W| \cdot \lambda}{\sum_{w \in W} w} + (1 - \lambda),$$

where W is the multiset containing each w_x and λ is the regularization parameter.

When $\lambda = 1$, each segment contributes equally to the loss across the entire training process. Conversely, when $\lambda = 0$, each segment contributes equally to the loss of a single datapoint.

The loss of a single datapoint was calculated by multiplying the loss of each segment x with the normalized weight w'_x before summing them up as before. We trained models with regularization parameters 0.1, 0.2, 0.3, 0.4, 0.7, and 1.

We only used 3/5 seeds (55555, 66666, and 77777). This was because we could guess from initial experiments that this would not lead to any significant improvement.

3.2 Pretraining a Masked Autoencoder for Feature Extraction (Pretrained U-Net)

To have our segmentation model learn more robust representations that do not rely solely on labels, we experimented with pre-training a Masked Autoencoder (MAE) and then use the encoder part for extracting features within our segmentation model. Doing this also allowed us to make use of the large amounts of unlabeled data available in our in-house dataset.

Dataset

The data available for training the MAE consisted of 2579 videos which contained a total number of 506'133 frames. The videos did not contain any patient metadata, so it is unclear how many patients are included. The actual dataset used for training only contains every 5th frame from every video, making for a total of 102'250 frames in the dataset. This was done to reduce the size of the dataset for practical reasons.

The dataset was split into 20% test set, 64% training set, and 16% validation set. Because patient information was not available, it is possible that videos from the same patient appear in training, validation, and test set. As we only care about the performance of the downstream task and not the MAE itself, this is acceptable.

Preprocessing

All images were resized to 384x256 using inter-area interpolation (through the OpenCV library) and converted to grayscale. No padding was added, so it is possible that the original aspect ratio of some images was not preserved. Since we vary the aspect ratio in later augmentations anyway, this does not matter too much.

Images from the training were further augmented using the Albumentations [30] library (version 1.4.11). Each time an image was used during training, it was augmented using RandomResizedCrop. This crops out a part of the image with a size between 30% and 100% (inclusive) of the original image and an aspect ratio between 0.75 and 1.33, before resizing the cropped part back to 384x256. The idea was to get the model to learn features that are robust to variations in scale and aspect ratio.

Images from the training, validation, and test set were randomly masked each time they were used. The image would get divided into patches of

3.2. Pretraining a Masked Autoencoder for Feature Extraction (Pretrained U-Net)

32x32, and then 75% of patches are masked at random. The masked patches would get replaced by a learnable mask token. This follows the masking strategy that was used by He et al. [28].

Architecture

The model architecture for the MAE is almost identical to the baseline segmentation model (see Figure 3.1 for an overview). The only difference lies in the last convolution layer that produces the final output. Compared to the segmentation model whose output has 4 channels, one for each predicted segment, the MAE model only outputs one channel that contains the reconstructed grayscale image.

Training

One training epoch consisted of 2'500 batches with a batch size of 8. This made for a total of 20'000 samples seen per training epoch. Gradients were accumulated for 8 steps, making for an effective batch size of 64. The samples of a training epoch were chosen at random, while ensuring at least 10 frames from every video in the training set were included.

A validation round was performed after every training epoch. One validation round consisted only of a subset of the entire validation set. 4000 samples would get chosen at random, while also ensuring that at least 10 frames from every video in the validation set were included. Training continued until the validation loss did not improve for 100 epochs.

The loss used for the reconstruction error was a pixel-wise mean squared error (MSE) loss. No loss was computed for visible patches. When adding a loss on the visible patches, we observed that the model would learn the identity function.

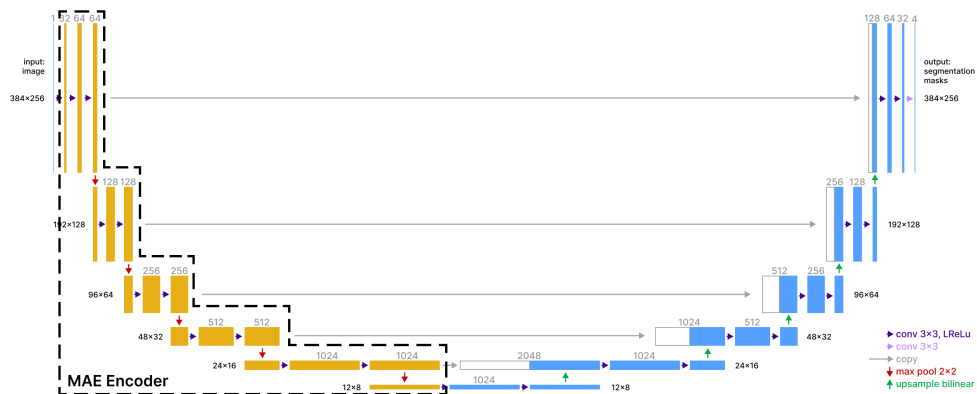
Much like with the segmentation model, we also employed a deep supervision loss on the second to last 3 output layers. See the training section of chapter 3.1 for a more detailed description.

For learning rate optimization we used AdamW [34] initialized to $5e^{-4}$ with a cosine decay scheduler with the minimum learning rate set to $1e^{-6}$. We further used a warmup period of 7'500 optimization steps. One optimization step corresponds to 8 training steps because we accumulate gradients over 8 training batches.

Evaluation

During training reconstructed images from validation rounds were logged for visual inspection.

The training of this model happened almost identically to that of the baseline model (see chapter 3.1). The major difference was that the encoder was initialized with the parameters obtained from the MAE encoder. We initially only trained these models on one of the seeds, namely 55555. The best performing models were then trained on the same 5 seeds as the baseline model to have a reasonable comparison.



analyze ultrasound videos. This would then ideally lead to an increase in segmentation quality, especially in otherwise hard-to-tell cases. To test this, we at first tried to keep the model architecture the same and would pass context frames via the channel dimension.

In a later experiments, we then also made use of 3D convolutions to extract temporal information. See chapters 3.4 and 3.5 for that.

Dataset

The dataset used for this and the following experiments is almost the same as for the baseline model. See the dataset section of chapter 3.1 for a detailed description.

It differs in two minor ways: firstly, because of an oversight by the author that was noticed too late, the dataset used here contains more videos (363 instead of 317) and more labeled frames (1700 instead of 1550). Luckily, the dataset statistics remain roughly the same, which is also why a detailed breakdown as in chapter 3.1 is omitted.

Secondly, because we require additional frames to pass to the model for context, all frames preceding any labeled frames were included when building this dataset.

Preprocessing

For each labeled frame, we also loaded $n - 1$ preceding frames that would be spaced s frames apart. These preceding frames were then added to the original input in separate channels. Both n and s were set as hyperparameters.

To give an example, let $n = 5$ and $s = 5$. For a labeled frame with index $i = 30$, we would additionally load frames with indices in $I_{preceding} = \{25, 20, 15, 10\}$. The final input would then have 5 channels, with each channel containing frame 30, 25, 20, 15, and 10 respectively. All images were in grayscale, which is why 1 channel per image is enough.

In the case of e.g. $i = 8$, where we would have $I_{preceding} = \{3, -2, -7, -12\}$, frames with negative indices were replaced with the labeled frame. In this case that would mean each 4 channels each contained frame 8, and 1 channel contained frame 3.

All other preprocessing was done as with the baseline model. See the preprocessing section of chapter 3.1. It was ensured that all frames of the same input would be augmented in the exact same way.

Architecture

The architecture used is almost identical to the architecture of the baseline model, with the only difference being that the input already contains 16 or 32 channels before the initial convolution. See Figure 3.1 for an overview of the architecture.

Training

We trained models with input sequences of length $n \in \{16, 32\}$, and strides $s \in \{1, 2, 5\}$. Other than that, training was done as with the baseline model. See the training section of chapter 3.1.

Evaluation

Evaluation was done as with the baseline model. See the evaluation section of chapter 3.1.

3.4 3D Convolution on Image Sequences (3D U-Net)

When using 2D convolutions using multiple frames by means of channels, the model has no way of knowing how the frames relate temporally. For this, we needed to make use of 3D convolutions, where the kernel would also slide along the temporal dimension in addition to the image dimensions.

Dataset

The dataset used is the same as with the previous experiment. See the dataset section of chapter 3.3.

Preprocessing

Preprocessing was identical to what was described in chapter 3.3, except that the preceding frames were put into an additional temporal dimension instead of channels. See the preprocessing section of chapter 3.3 for more detail.

Architecture

For this experiment we had to adapt the baseline architecture to work with 3D convolutions. As in Çiçek et al. [26], we replaced all 2D operations with their respective 3D variants. This meant 2D convolutions became 3D convolutions, 2D max pooling became 3D max pooling, and bilinear upsampling became trilinear upsampling.

3.4. 3D Convolution on Image Sequences (3D U-Net)

Because we only had labels for the last frame of an input and not for the preceding frames, we added one final convolution which reduced the size of the temporal dimension to 1. See Figure 3.3 for an overview of the model.

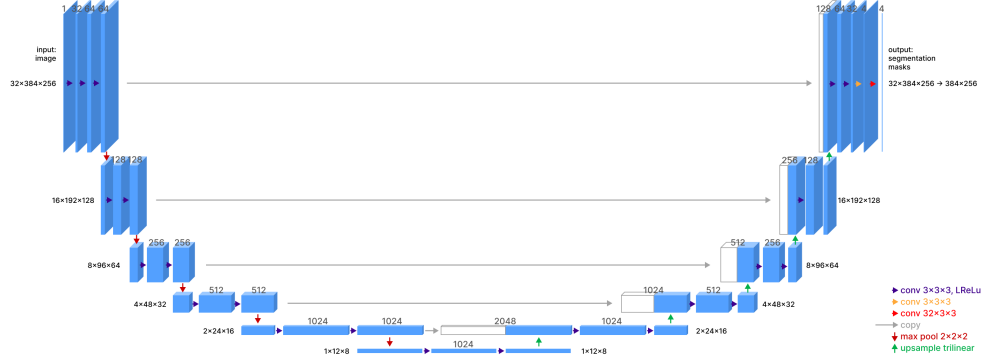


Figure 3.3: 3D U-Net architecture. The overall structure is identical to the 2D U-Net depicted in Figure 3.1. An additional temporal dimension is added. The last convolution has a kernel that spans the entire temporal dimension and reduces its size to 1.

We also experimented with a shallower version of the 3D U-Net that had 4 layers instead of 5 to reduce model complexity.

Training

The training loop again remained mostly identical to that of the baseline model (see the training section of chapter 3.1). The main difference was that the batch size had to be decreased to 1, as the GPU kept running out of memory otherwise. Batch normalization now no longer made sense, so we switched to instance normalization. To keep the effective batch size at 8, we accumulated gradients over 8 training steps.

We trained the model on input sequences of length $n \in \{16, 32\}$, and strides $s \in \{2, 5\}$.

Based on initial trial runs, we knew that the 3D U-Net would take a long time to train (over 25 hours) and likely not improve segmentation quality. Due to that, we decided to only train each model variation on only 2/5 seeds. Those were 55555 and 77777, as they were the best respectively worst performing seeds with the baseline model.

Evaluation

Evaluation was done the same way as with the baseline model. See the evaluation section of chapter 3.1.

3.5 Hybrid 2D/3D Convolution on Image Sequences (2.5D U-Net)

As only the last frame within a given sequence would be labeled with ground truth masks, we only ever calculated a loss on that last frame. It seemed inefficient to potentially produce segmentation masks for all images in a sequence when all but the last would provide the model with any feedback. The increased model complexity could also lead the model to overfit the training data quicker.

This led us to a hybrid approach where we made use of both 2D and 3D convolutions. Early encoder layers would use 3D convolutions to extract both temporal and spatial features. In later encoder and decoder stages, the model transitions to 2D convolutions. This keeps both model complexity and training time low compared to the full 3D U-Net.

Dataset

The same dataset as with the other experiments on image sequences was used. See the dataset section of chapter 3.3 for a description.

Preprocessing

Preprocessing was identical to what was described in chapter 3.3, except that the preceding frames were put into an additional temporal dimension instead of channels. See the preprocessing section of chapter 3.3 for more detail.

Architecture

The model features 3D convolutions in the first 3 layers of the encoder. After that, the temporal dimension is collapsed by dropping all but the last dimension, corresponding to the labeled frame. The remaining encoder and decoder layers all use 2D convolutions. To facilitate the skip connections between the 3D layers of the encoder and the 2D layers of the decoder, only the last temporal dimension is copied over.

Training

Just like with the trained models with input sequences of length $n \in \{16, 32\}$, and strides $s \in \{1, 2, 5\}$. Other than that, training was done as with the baseline model. See the training section of chapter 3.1.

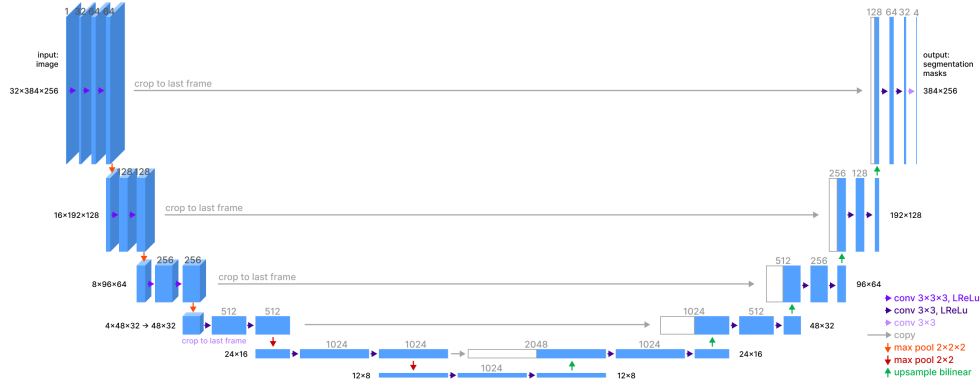


Figure 3.4: Hybrid 2D/3D U-Net architecture. After the third encoder layer, all but the last frames are dropped, and the model transitions from 3D convolutions to 2D convolutions. This frame-dropping also happens in the skip connections between 3D encoder and 2D decoder layers.

Evaluation

Evaluation was done the same way as with the baseline model. See the evaluation section of chapter 3.1.

3.6 Addressing Determinism in Experiments

In order to have our results be reproducible and useful in comparisons, we set all global seeds of any library used. PyTorch and the CUDA backend may per default use non-deterministic algorithms for performance reasons. We ensured that deterministic algorithms were enforced whenever possible. Only for models that included 3D convolutions was this not possible, as no deterministic version of the 3D max pool operation existed in the CUDA backend. Any variance introduced by this should be rather small though.

Despite all this, training the same model twice with the exact same parameters still yielded varying results, with more variance than can be explained away solely by the use of non-deterministic CUDA algorithms. Unfortunately, we were unable to figure out what caused this. To bound the variance of our results, the baseline model as well as any model that showed promise was evaluated multiple times. Through this, even if results may not be exactly reproducible, general statements about the relative performance of different models can be made.

Chapter 4

Results

First, different variations of models from the same experiments are compared to each other and the baseline by their achieved Dice score. After that, we show a comparison of the best models from each experiment and also take a look at the frame-level sensitivity and specificity of those model. Unless stated otherwise, all Dice scores were computed only on frames containing the segment in question, i.e. positive samples.

At the end, a qualitative comparison of the best performing models is presented.

4.1 2D Convolution on Single Frames (Baseline U-Net) And Adjusted Loss Weights

Adjusting the weights of losses did not seem to meaningfully improve on the baseline model. The segmentation performance on all segments generally goes down with stronger regularization. A regularization parameter of 0.3 performs slightly better than the baseline, but it seems to be an outlier. See Figure 4.1.

4.2 Pretraining a Masked Autoencoder for Feature Extraction

The Masked Autoencoder trained for 2 days and 22 hours before the early stopping condition would cause training to stop. After the first 150 epochs of MAE training, the downstream task shows little to no improvement. Downstream models all either performed better or equal to the baseline (see Figure 4.2). Performance improved when the pretrained encoder was further finetuned during the downstream task (see Figure 4.3).

4. RESULTS

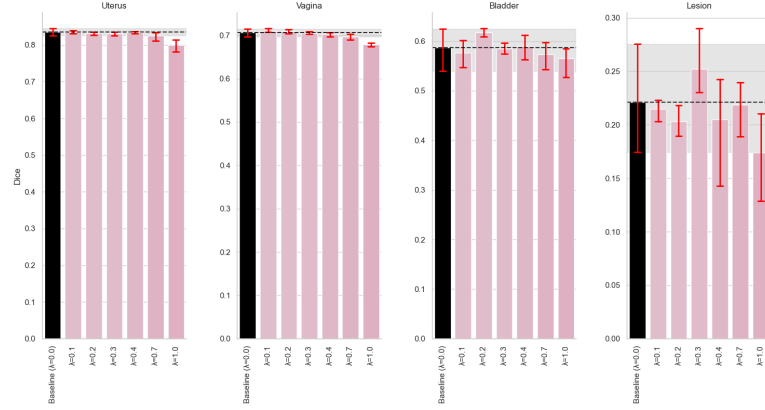


Figure 4.1: Dice scores of the baseline and weight adjusted models across all segments. Note that each subgraph has its own scale. The black bar represents the baseline model. The pink bars were models trained with an increasing loss weight regularization parameter λ . The black dotted line shows the mean dice of the baseline model for easier comparison. The red error bars show the 95 percentile interval. The lower end is the 2.5th percentile, while the upper end is the 97.5th percentile. The gray backdrop shows this area for the baseline model for easier comparison.

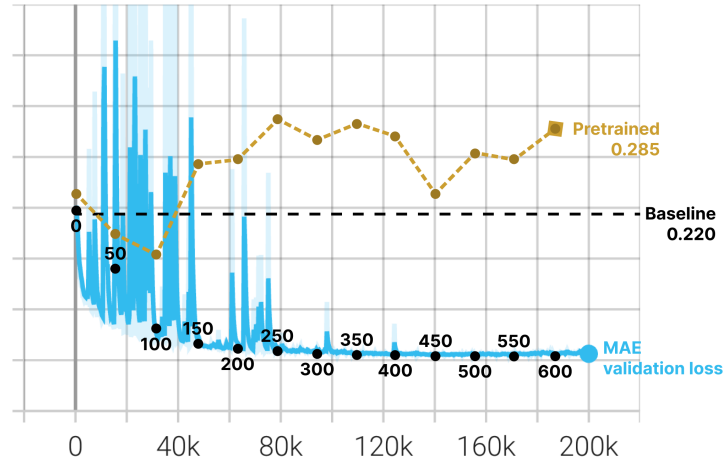


Figure 4.2: The solid blue line shows the validation loss of the MAE over 600 epochs. The loss starts at 0.030 and goes down towards 0.015. The numbers at the bottom show optimization steps. The black dots show the points in epoch time when a downstream model was trained with the MAE encoder. The dashed yellow line shows the evolution of the Dice score of the downstream tasks on lesions. The score of the last downstream task is shown. The dashed black line shows the Dice score of the baseline model on lesions. The y-axis ticks, which showed the MSE loss, were removed to prevent confusion with the Dice scores.

4.3. 2D Convolution on Image Sequences

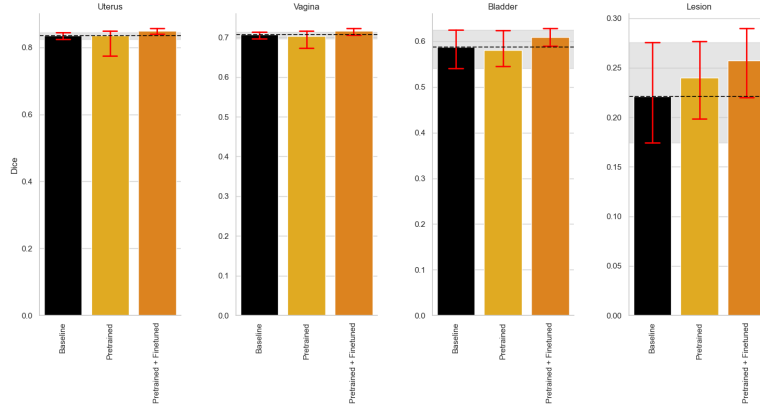


Figure 4.3: Dice scores of pretrained models across all segments. Note that each subgraph has its own scale. The black bar represents the baseline model. The lighter yellow bar shows the mean of downstream models where the encoder was frozen during training. The orange bar shows the mean of models trained on the same pretrained encoders, but they were allowed to be finetuned. The black dotted line shows the mean dice of the baseline model for easier comparison. The red error bars show the 95 percentile interval. The gray backdrop shows this error area of the baseline model for easier comparison.

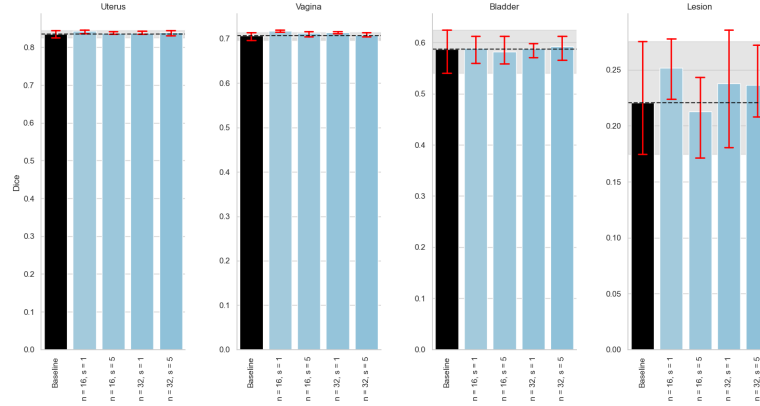


Figure 4.4: Dice scores of Channel-3D U-Nets across segments. Two different sequence lengths $n \in \{16, 32\}$ and strides $s \in \{1, 5\}$ made for a total of 4 trained models. Note that each subgraph has its own scale. The black bar represents the baseline model. The light blue bars represent different Channel-3D U-Net models. The label denotes the corresponding parameters n and s . The black dotted line shows the mean dice of the baseline model for easier comparison. The red error bars show the 95 percentile interval. The gray backdrop shows this error area of the baseline model for easier comparison.

4.3 2D Convolution on Image Sequences

The Channel-3D U-Net did not appear to improve the baseline performance by a significant amount. The overall mean for lesions is slightly higher, but not yet outside of the expected baseline range. See Figure 4.4.

4. RESULTS

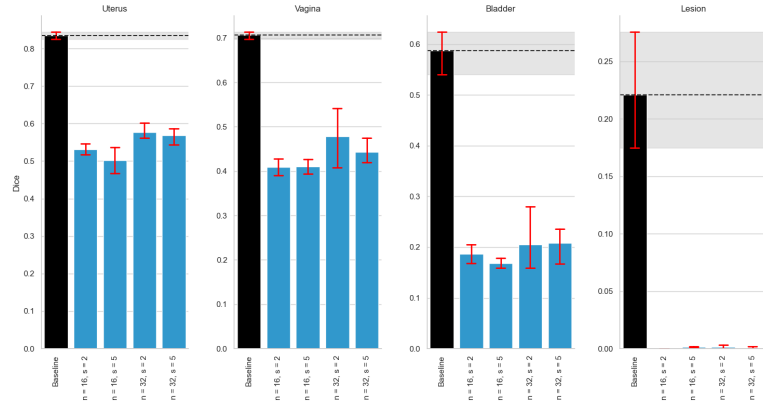


Figure 4.5: Dice scores of 3D U-Net models across segments. Each subgraph has its own scale. The black bar is the baseline model. The light blue bars are different Channel-3D U-Net models. The label denotes the corresponding parameters n and s . The values of models that only differed in their layer count were combined to keep the graphs simple. The black dotted line shows the mean dice of the baseline model. The red error bars show the 95 percentile interval. The gray backdrop shows this error area of the baseline model.

4.4 3D Convolution on Image Sequences

While initial 3D experiments had already performed significantly worse than the baseline, when rerunning training with the same seeds as all other experiments, the model seemed to start overfitting on the training set quicker than usual. This led to an early stopping of training, and resulted in very bad test scores. See Figure 4.5.

4.5 Hybrid 2D/3D Convolution on Image Sequences

Similarly to the Channel-3D U-Net, the overall mean of the 2.5D U-Net for lesions is slightly above the mean of the baseline, but still within bounds. The variance within a single model has seemingly decreased though. See Figure 4.6.

4.6 Overall comparison

The pretrained U-Net delivered the best results in terms of Dice score. For 3 out of 4 segments, including lesions, it achieved the highest values out of all models examined. For lesions, this meant an improvement of 18.55% over the baseline. All other segments only saw improvements of 1% to 2%.

Frame-based and pixel based sensitivity also saw improvements of 18.15% respectively 19.23% with the pretrained U-Net compared to the baseline model.

4.6. Overall comparison

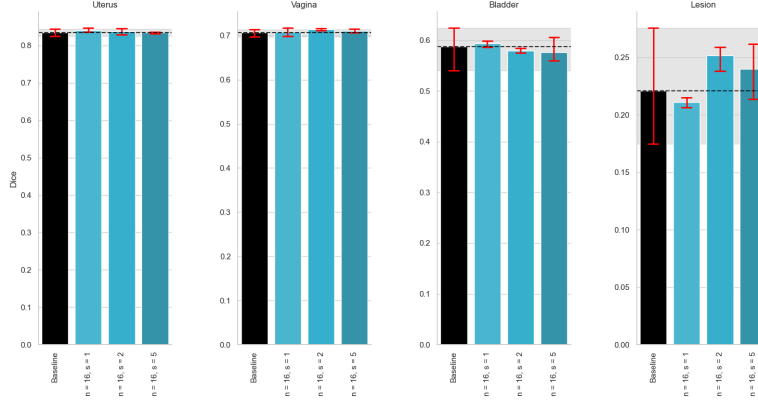


Figure 4.6: Dice scores of the 2.5D U-Net across segments. The differing model parameters n and s are shown below each bar. Note that each subgraph has its own scale. The black bar represents the baseline model. The black dotted line shows the mean dice of the baseline model for easier comparison. The red error bars show the 95 percentile interval. The lower end is the 2.5th percentile, while the upper end is the 97.5th percentile. The gray backdrop shows this area for the baseline model for easier comparison.

Table 4.1: Mean Dice scores per segment for each model. See Figure 4.7 for a visual display.

	Uterus	Bladder	Vagina	Lesion
Baseline	0.835	0.587	0.706	0.221
Pretrained U-Net	0.854	0.593	0.714	0.262
Channel-3D U-Net	0.843	0.589	0.717	0.252
3D U-Net	0.577	0.205	0.478	0.002
2.5D U-Net	0.834	0.576	0.711	0.240

Table 4.2: Sensitivity and Specificity of Lesion Detection and Segmentation

	Pixel-based Segmentation		Frame-based Detection	
	Sensitivity	Specificity	Sensitivity	Specificity
Baseline	0.026	0.999	0.484	0.958
Pretrained U-Net	0.031	0.998	0.572	0.963
Channel-3D U-Net	0.028	0.999	0.513	0.966
3D U-Net	0.000	1.000	0.010	0.991
2.5D U-Net	0.026	0.999	0.482	0.972

4. RESULTS

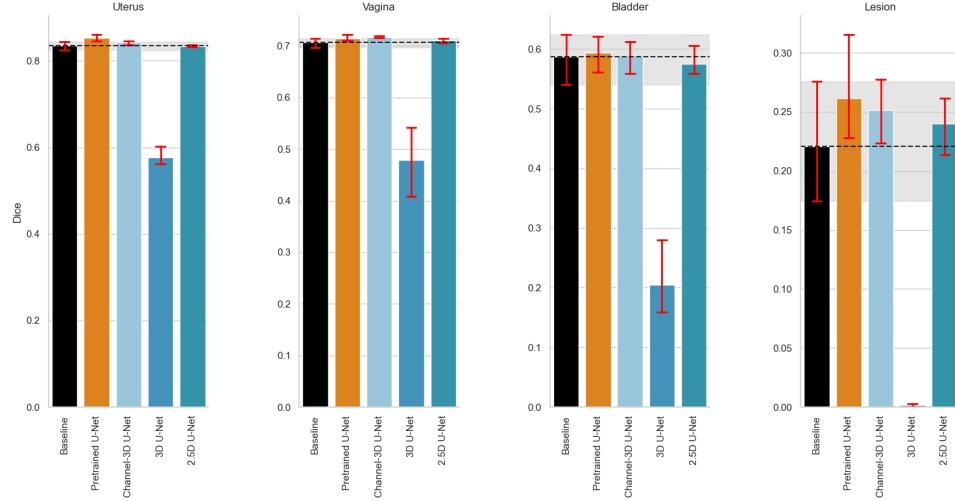


Figure 4.7: Comparison of the Dice scores of the best models from each experiment. From left to right: the black bar is the baseline, orange is from the pretrained (+ finetuned) U-Net, light blue is the Channel-3D U-Net, blue is the full 3D U-Net, and turquoise is the 2.5D U-Net. The dashed black line shows the mean Dice score of the baseline. The red error bars show the 95 percentile interval

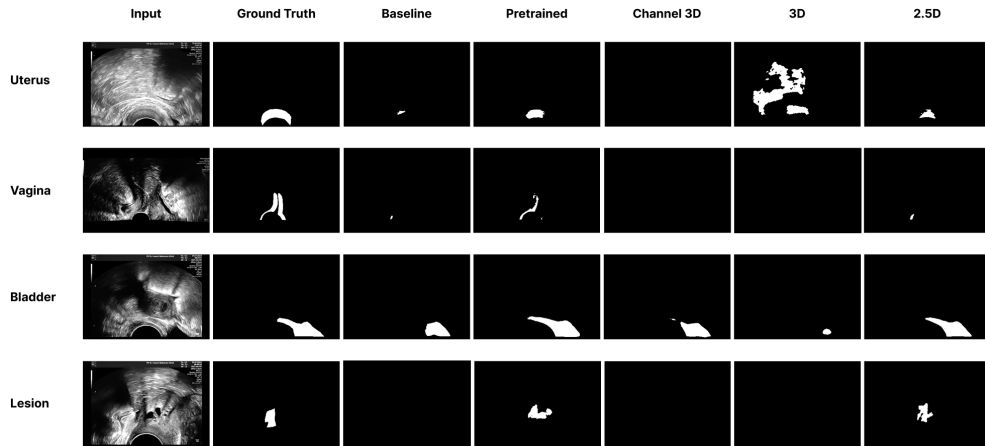


Figure 4.8: Qualitative comparison of predicted segmentation masks across models.

Discussion and Conclusion

This thesis explored various deep-learning-based approaches to improving the segmentation of pelvic organs and endometriotic lesions in ultrasound images and videos. The primary goal was to enhance the performance of a baseline U-Net model developed by Scanvio by incorporating additional spatial and temporal information as well as leveraging unsupervised pre-training methods.

5.1 Summary of Key Findings

The most significant finding was that pretraining a Masked Autoencoder (MAE) and using the pretrained encoder for feature extraction led to the highest improvement in segmentation quality. This approach improved the Dice score for lesion segmentation by 18.55% compared to the baseline. Additionally, sensitivity and specificity for lesion detection also showed marked improvements. Finetuning the pretrained encoder further enhanced the model's performance.

Other explored methods, such as using 2D convolution with multiple frames (Channel-3D U-Net), 3D convolution (3D U-Net), and a hybrid 2.5D approach, did not provide significant improvements. The Channel-3D U-Net and 2.5D U-Net showed slight gains in lesion segmentation, but the improvements were within the expected variance of the baseline model. Further experimentation would be required, especially with the 2.5D U-Net, before a conclusive statement can be made. The full 3D U-Net, however, performed significantly worse, likely due to overfitting and increased computational complexity.

5.2 Limitations and Challenges

The dataset was relatively small, with only 1550 labeled images across 317 videos. This may have limited models in learning to generalize well, especially for the least frequent class which was lesions (13.1% of dataset images).

The unexplainable variance between identical training runs makes it harder to properly trust when a certain model performs particularly well, as even the base model had quite high variance within different runs.

The lack of a temporal ground truth when training the 3D U-Net definitely played a role in its poor performance. Having more labeled frames closer together would allow for guiding the model better during training.

Due to the model complexity and resulting hardware limitations, the 3D U-Net could also not be explored fully, as only a handful of experiments could be ran with only very small batch sizes (i.e. 1).

An additional challenge for the 3D U-Net approach was that videos would have differing framerates as well as differing movement speeds. Depending on the movement of the operator of the ultrasound probe or the patient, adjacent frames could sometimes look wildly different. Other times, frames would look almost the same over the course of a hundred frames. This made it hard to decide on a good context size (i.e. how many frames to pass and how apart should they be).

5.3 Future Research Directions

Based on the success of the Masked Autoencoder approach, and inspired by Feichtenhofer et al. [?], it would be interesting to train a 3D MAE which could be combined with either a hybrid 2D/3D model or a full 3D model to reduce training times by providing a good initialization.

A further idea not pursued during this thesis were to define a temporal consistency loss on frames of a 3D input sequence, such that the predictions on frames that did not contain a label had to at least somewhat be similar to their neighboring frames. One could start with the second to last frame, for which a neighboring ground truth exists, and then work backwards, reducing the loss weight the further back frame is.

It might also be interesting to explore different kinds of encoder architectures apart from just CNNs. As demonstrated by He et al. [28], transformers can allow for much more efficient training when the masking ratio is high compared to CNNs.

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